



Vincenzo Gulisano, Ph.D.
Assistant Professor

Distributed Computing and Systems Research Group.
Chalmers University of Technology
Gothenburg, Sweden

Date of birth: 15/04/1984
Nationality: Italian

My research focuses on scalable big data analysis in cyber-physical systems such as Smart Grids, Advanced Metering Infrastructures and Vehicular Networks.

During my Ph.D., I designed and developed StreamCloud: a pioneer parallel, distributed and elastic Stream Processing Engine. I am currently investigating streaming operators' parallelization, load balancing, provisioning, decommissioning and fault tolerance protocols. At the same time, I am also looking into enhancement of parallel stream operators by means of streaming- and energy-aware concurrent data structures that can significantly boost analysis performance (both in throughput and latency). I also research applied use of the streaming paradigm in data validation, intrusion detection systems, privacy-preserving online analysis, distributed embedded networks and cloud infrastructures.

Research Lines

- Data Streaming
 - Parallel/Concurrent Data Structures
 - Multi-core and multi-node architectures
 - Distributed Systems
 - Security in Advanced Metering Infrastructures
 - Smart grids
 - Load Balancing, Elasticity and Fault Tolerance protocols
 - DDoS Detection / Mitigation
 - Network Analysis
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Education

Ph.D. Degree

Universidad Politécnica de Madrid, Spain. December 2012

Master's Degree in Computer Engineering

Università degli Studi di Trieste, Italy. July 2008

Bachelor's Degree in Computer Engineering

Università degli Studi di Trieste, Italy. July 2006

Teaching and pedagogical experiences

Teaching experience

- **Operating Systems.** Chalmers University of Technology. December 2015.
- **Operating Systems.** Chalmers University of Technology. December 2014. 3rd year course, 119 students registered.
- **ICT Support for Adaptiveness and (Cyber)security in the Smart Grid.** Chalmers University of Technology. April 2014 (Developed the material for a lecture about Data Streaming). Master course.

- **Operating Systems.** Chalmers University of Technology. December 2013 (developed the material and taught a set of lectures). 3rd year course, 112 students registered.
- **ICT Support for Adaptiveness and (Cyber)security in the Smart Grid.** Chalmers University of Technology. April 2013 (helped in the organization of the course, followed students' projects, and developed the material for a lecture about Data Streaming). Master course.
- **Algorithms and Data Structures.** Technical University of Madrid. September 2011 (developed the material and taught a set of lectures). 2nd year course, 80 students registered.

Supervision of Master Thesis

- Applied Differential Privacy in the Smart Grid. Hedvig Jonsson & Boel Nelson. 2015
- Distributed and Online Advanced Metering Infrastructures Data Validation using Single-Board Devices. Jonas Sandström. 2015
- Visualizing HTTP traffic flows from packet data. Shailaja Mallick. 2015
- BCStream – a data streaming based system for processing energy consumption data and integrating with social media. Carl Wälinder & Bao Hoang. 2014 (Undergoing)
- Know your neighbor: self-organizing data streaming processing in Advanced Metering Infrastructures. Theodoros Kotsantinis. 2014
- Robustness testing using model-based fuzzing. William Johansson and Martin Svensson. 2013 (resulted also in a publication at the 7th International Conference on Software Testing, Verification and Validation (ICST) 2014)

Supervision of Undergraduate projects

- The Smart Power Strip. Bregell Johan, Hillbon Erik, Johansson David, Karlsson Philip, Larsson Jacob, Stigelid Marcus (awarded with highest grade by the examiner).
- The Smarter Power Strip – Design and implementation of a smart power strip. Marco Baxemyr, Johannes Broberg, Karl Buchka, Mattias Henriksson, Gustav Olsson, Johan Swetzen (awarded with highest grade by the examiner).

Courses (participated)

Supervision of Research (CIU 930, Chalmers University of Technology).

Seminars and Courses (held by me):

- **An overview of Data Streaming and Stream Processing Engines.** Distributed Systems Group, INESC-ID. Lisbon. Portugal. February 2013.
 - **An overview of Data Streaming and Stream Processing Engines.** Distributed Computing and System Research Group, Chalmers University of Technology. November 2011.
 - **Data Stream Management Systems.** Ericsson Spain, Madrid R&D Center, Technology & Innovation. May 2010 (6 lessons).
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Research Experience

- **Distributed Computing and System Research Group**
Chalmers University of Technology.
June 2012 and November 2011
 - **Institute of Computer Science (ICS)**
Foundation for Research & Technology (FORTH).
December 2010
 - **Microsoft Summer School**
Cambridge, UK. June 2009.
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Publications

1. **Data-Streaming and Concurrent Data-Object Co-design: Overview and Algorithmic Challenges.**
Vincenzo Gulisano, Yiannis Nikolakopoulos, Marina Papatriantafilou, Philippas Tsigas. Book: Algorithms, Probability, Networks, and Games. Springer International Publishing, 2015.
2. **ScaleJoin: a deterministic, disjoint-parallel and skew-resilient stream join.**
V. Gulisano, Y. Nikolakopoulos, M. Papatriantafilou, P. Tsigas. IEEE International Conference on Big Data (IEEE Big Data 2015)
3. **STONE: A Stream-based DDoS Defense Framework**
Vincenzo Gulisano, Mar Callau-Zori, Zhang Fu, Ricardo Jiménez-Peris, Marina Papatriantafilou, Marta Patiño-Martínez Journal: Expert Systems with Applications. 2015
4. **DEBS Grand Challenge: Deterministic Real-Time Analytics of Geospatial Data Streams through ScaleGate Objects.**
Vincenzo Gulisano, Yiannis Nikolakopoulos, Ivan Walulya, Marina Papatriantafilou, Philippas Tsigas. The 9th ACM International Conference on Distributed Event-Based Systems (DEBS 2015) [*Best Grand Challenge Solution Award*]
5. **When Smart Cities meet Big Data**
Vincenzo Gulisano, Magnus Almgren and Marina Papatriantafilou. **Journal:** ERCIM News No. 98 published - Special theme: Smart Cities
6. **Online and Scalable Data Validation in Advanced Metering Infrastructures**
Vincenzo Gulisano, Magnus Almgren, Marina Papatriantafilou. **Conference:** The 5th IEEE PES Innovative Smart Grid Technologies (ISGT) European 2014 Conference
7. **METIS: a Two-Tier Intrusion Detection System for Advanced Metering Infrastructures**
V. Gulisano, M. Almgren, M. Papatriantafilou. **Conference:** 10th International Conference on Security and Privacy in Communication Networks (SecureComm) 2014
8. **[BRIEF ANNOUNCEMENT] Concurrent Data Structures for Efficient Streaming Aggregation**
D. Cederman, V. Gulisano, Y. Nikolakopoulos, M. Papatriantafilou, P. Tsigas. **Conference:** The 26th Annual ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2014
9. **METIS: a Two-Tier Intrusion Detection System for Advanced Metering Infrastructures**
V. Gulisano, M. Almgren, M. Papatriantafilou. **Conference:** The fifth International Conference on Future Energy Systems (ACM e-Energy) 2014.
10. **T-Fuzz: Model-Based Fuzzing for Robustness Testing of Telecommunication Protocols**
W. Johansson, M. Svensson, U. Larson, M. Almgren, V. Gulisano. **Conference:** 7th International Conference on Software Testing, Verification and Validation (ICST) 2014
11. **A Scalable SIEM Correlation Engine and its Application to the Olympic Games IT Infrastructure**
V. Vianello, V. Gulisano, R. Jimenez-Peris, M. Patiño-Martínez, R. Torres, R. Díaz, E. Prieto. **Conference:** 8th International Conference on Availability, Reliability and Security (ARES), 2013
12. **Algorithms and Data Handling Towards Adaptive and Robust Electricity Networks**
M. Almgren, D. Cederman, Z. Fu, V. Gulisano, O. Landsiedel, M. Papatriantafilou. **Conference:** Chalmers Energy Conference 2013
13. **Cybersecurity in the Smart Grid**
M. Almgren, Z. Fu, V. Gulisano, M. Papatriantafilou, V. Tudor. **Conference:** Chalmers Energy Conference 2013
14. **STONE: A Stream-based DDoS Defense Framework**
Mar Callau-Zori, Vincenzo Gulisano, Zhang Fu, Ricardo Jiménez-Peris, Marina Papatriantafilou, Marta Patiño-Martínez **Conference:** Proceedings of the 28th Annual ACM Symposium on Applied Computing, SAC 2013
15. **Distributed and Scalable SIEM for the Olympic Games Infrastructure**
Valerio Vianello, Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Rubén Torres, Rodrigo Díaz, Elsa Prieto **Conference:** XXI Jornadas de Concurrencia y Sistemas Distribuidos, JCSD 2013

16. **StreamCloud: An Elastic and Scalable Data Streaming System**

Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Claudio Soriente, Patrick Valduriez **Journal:** IEEE Transactions on Parallel and Distributed Systems, TPDS 2012

17. **A Big Data Platform for Large Scale Event Processing**

Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Claudio Soriente, Patrick Valduriez
Journal: ERCIM news, April 2012

18. **Complex Event Processing Based SIEM**

Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Claudio Soriente, and Valerio Vianello.
Book: Advances in Security Information Management: Perceptions and Outcomes, 2011

19. **Elastic Parallel Data Streaming**

Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Claudio Soriente, Patrick Valduriez
Conference: Jornadas de Concurrencia y Sistemas Distribuidos, JCSD 2011

20. **StreamCloud: A Large Scale Data Streaming System**

Vincenzo Gulisano, Ricardo Jiménez-Peris, Marta Patiño-Martínez, Patrick Valduriez **Conference:** IEEE 30th Int. Conf. on Distributed Computing Systems, ICDCS 2010

21. **A New Class of Services for Cloud Computing: Real-Time Services over Massive and Continuous Data Flows**

Ricardo Jiménez-Peris, Marta Patiño-Martínez, Vincenzo Gulisano **Conference:** Cloud Futures 2010 Workshop, 2010

Participation in Research Projects with Public Funding

- **EXAMINE: Extracting useful information out of data in AMI networking.** Göteborg Energi, Chalmers Area of Advance
 - **CRISALIS: CRITICAL Infrastructure Security AnaLysis**
7th FP European Commission, 2012-2015
 - **SysSec Consortium**
7th FP European Commission, 2010-2013.
 - **MASSIF: Management of Security Information and events in Services Infrastructures**
7th FP European Commission, 2010-2013
 - **IOLanes: Advancing the Scalability and Performance of I/O Subsystems in Multi-core Platforms**
7th FP European Commission, 2010-2013
 - **CLOUDS: Cloud Computing para Servicios Escalables, Confiables y Ubicuos**
Comunidad Autónoma de Madrid, 2010-2013
 - **CloudStorm: Scalable and Dependable Cloud Service Platforms**
Ministry of Science and Innovation (MICINN), 2010-2013
 - **Stream: Scalable Autonomic Streaming Middleware for Real-time Processing of Massive Data Flow**
7th FP European Commission, 2008-2011
 - **Highly Scalable Platform for the Construction of Dependable and Ubiquitous Services**
Ministry of Science and Innovation (MICINN), 2007-2010
 - **PubSub4RT Publish Subscribe for Real Time Services.**
EUREKA/ITEA-AVANZA I+D, 2008-2009
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Other merits – Awards

- Best Grand Challenge solution award for paper “Deterministic Real-Time Analytics of Geospatial Data Streams through ScaleGate Objects” at DEBS 215
 - Grants from the EU network of excellence SysSec for two visits at Chalmers University (June 2012 and November 2011)
 - **Ph.D. Fellowship** Ayudas para la realización del Doctorado en sus Departamentos, Centros I+D e Institutos (Universidad Politécnica de Madrid)
 - **Microsoft PhD Scholarship Programme**
 - **Microsoft Student Partner for the University of Trieste** Academic Year 2007-2008.
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Personal skills and competences

- **Programming languages:** C++ (2008-1012), Java (2008-present). Others: C#, Python, Visual Basic, XML/XSD/XSLT manipulation, SQL, Perl, Bash scripting, Aspx web programming.
 - **Others:** Storm, Hadoop File System, Orange File System, Matlab, Project management with Autotools, Versioning with GIT and SVN repositories.
 - **IDEs:** Visual Studio, NetBeans, Eclipse.
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Languages

- **Italian:** mother tongue
 - **English:** C2. Proficient user
 - **Spanish:** C2. Proficient user
 - **Swedish:** A1. Breakthrough / beginner
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